

LeadFairRec: LLM-enhanced Discriminative Counterfactual Debiasing for Two-sided Fairness in Recommendation

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Abstract

Fairness-aware recommendation has emerged as a pivotal research area in recent years. Current fairness studies primarily examine two independent dimensions: user-side fairness and item-side fairness. However, most approaches address each side's fairness in isolation while neglecting their complex interdependencies. In this paper, we propose an LLM-Enhanced Discriminative Counterfactual Debiasing Model for Two-sided Fairness in Recommendation (LeadFairRec). Specifically, we first design a two-sided causal graph that jointly models provider-customer fairness interactions through their causal relationships. Then we propose a discriminative counterfactual debiasing method, which effectively removes spurious correlations while maintaining true user-item interactions. Finally, we propose an LLM-enhanced counterfactual inference method to derive noise-resistant user/item representations from interaction data, enhancing the robustness of causal debiasing. The experimental results demonstrate the high effectiveness of our proposed model. We provide our code at <https://github.com/houyimin660/LeadFairRec>.

CCS Concepts

• **Information systems** → **Recommender systems**.

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Keywords

Recommender System; Two-sided Fairness; Discriminative Counterfactual Debiasing; LLM-Enhanced Counterfactual Inference

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1 Introduction

As recommender systems play an increasingly prominent role in the allocation of social resources, their fairness issues have attracted widespread attention. Current research on fairness in recommender systems primarily focuses on two dimensions: user-side fairness and item-side fairness. From the perspective of user fairness, recommender systems exhibit significant performance disparities across user groups. Active users consistently receive high-quality recommendations, while less active users face deteriorating service quality, resulting in a persistent recommendation gap. Regarding item fairness, the systems demonstrate a pronounced Matthew Effect: top sellers dominate recommendation slots, while long-tail items struggle to gain adequate exposure, often having their actual quality overlooked by the system. Fairness-aware recommendation holds significant importance for various applications such as e-business and content platforms. In e-commerce, ensuring equitable exposure opportunities for merchants of different scales not only helps consumers discover more quality products, but also promotes healthier platform ecosystem development. For content recommendation platforms, balancing recommendation quality between active and inactive users can enhance the overall user experience and increase

platform engagement. These wide applications have demanded fairness-aware recommendation.



Figure 1: Fairness guarantees for each side.

We study the problem of fairness-aware recommendation. Given a group of users and their historical interaction data, we need to address the unfair situation from both the user and the item (i.e., customer and provider) perspectives. For fairness-aware recommendation, three key issues need to be addressed. First, we need to design a recommendation model that balances user and item fairness. Simply combining fairness methods will not work due to inherent trade-offs between them. As shown in Figure 1, consider an active user Alice prefers niche brands such as OPPO. Standard user fairness methods would reduce her influence, but this also hurts niche brands' visibility. Similarly, enforcing item fairness for inactive users who prefer popular items may worsen user fairness. A well-designed fairness-aware recommendation model should effectively capture this two-sided trade-off to achieve multi-stakeholder fairness. Second, we need to adopt a nuanced approach toward biases in recommender systems, i.e., preserving their beneficial components rather than indiscriminately eliminating them. Take the user activity level as an example: while commonly regarded as a data bias, it inherently contains valuable behavioral signals. Active users typically provide more comprehensive and stable preference indicators. Similarly, item popularity, though potentially leading to the Matthew Effect, may genuinely reflect superior quality. A high-quality fairness-aware recommendation model should employ sophisticated debiasing. Third, robust representation learning is critical to model users/items accurately while filtering interaction noise. Real-world data frequently contains spurious patterns (e.g., accidental clicks on mismatched items like a tech fan browsing baby products). When these patterns occur frequently, they can lead to more severe consequences (e.g., baby products gaining artificial popularity through amplified error signals). An effective model must distinguish noise from true preferences by aligning user profiles with item attributes, thus maintaining recommendation reliability.

Recently, fairness in recommender systems has gained increasing attention in research, including single-sided [1, 2, 4, 5, 8, 12, 16, 21, 27, 31] and two-sided [13, 17, 20, 22, 23, 25] approaches. However, most studies focus on the fairness of each side independently and ignore their mutual influence, leading to superficial

fairness on one side at the expense of the other. Moreover, existing methods fail to consider the causal relationships underlying these complex interactions, resulting in spurious fairness adjustments that may inadvertently amplify biases. To overcome the problems of existing recommendation approaches, we propose an **LLM-Enhanced Discriminative Counterfactual Debiasing Model for Two-sided Fairness in Recommendation (LeadFairRec)**. Specifically, a two-sided causal graph is designed to jointly model the interactions between user and provider fairness. This causal graph integrates previously isolated fairness metrics (e.g., user parity, provider exposure) by explicitly capturing their interdependencies through causal relationships. Then, a discriminative counterfactual debiasing method is proposed to selectively preserve beneficial bias components while eliminating unfair influences. Finally, we leverage Large Language Models (LLMs) to analyze behavioral patterns and generate refined user/item representations that distinguish genuine preferences from random noise. We summarize our contributions as follows.

- (1) We design a two-sided causal graph that jointly models provider-customer fairness interactions through their causal relationships. With this graph, we can identify unfairness drivers from both perspectives while incorporating a personalized brand effect (*PBE*) that simultaneously affects users and providers.
- (2) We propose a discriminative counterfactual debiasing method. By maximizing the Total Indirect Effect (*TIE*) and strategically blocking confounding paths while preserving genuine signals, our approach effectively removes spurious correlations while maintaining true user-item interactions.
- (3) We propose an LLM-enhanced counterfactual inference method to derive noise-resistant user and item representations from interaction data, enhancing the robustness of causal debiasing. These semantically enriched profiles are integrated with the original user/item embeddings via contrastive learning, improving both the accuracy and fairness of recommendations.
- (4) We conduct extensive experimental evaluation on three real datasets. The experiment results demonstrate the high effectiveness of our proposed methods.

2 Related Work

In this section, we review the related work in three dimensions: user-side fairness, item-side fairness, and two-sided fairness in recommender systems.

User-side Fairness-aware Recommendation. Ensuring equitable treatment across user groups - measured through metrics like prediction accuracy [4] or explainability [5] - remains a core challenge in recommender systems. If recommendations are not fair to users, users with specific interests may be lost. The most general definition of fairness in user fairness is uniform fairness, because people generally believe that different people are similar and should not be treated differently. Current approaches employ various strategies: counterfactual interaction restructuring to mitigate group utility disparities [12], elimination of PS-Bias via Pearl's minimal-change principle [31], and context-aware distribution matching for paid scenarios [8]. However, these methods face two fundamental

limitations: they struggle with inherent fairness-utility trade-offs where fairness gains often compromise recommendation quality and critically overlook the dynamic nature of fairness requirements across both longitudinal user behavior patterns and individual equity needs.

Item-side fairness-aware recommendation. Item fairness examines whether recommender systems provide impartial treatment to different items, typically measured through metrics like comparable rating-prediction accuracy across item categories. If recommendations are unfair to different items, then the providers of these discriminated items may lack positive feedback and leave the platform. Recent research has focused on mitigating item-side popularity bias through three complementary approaches: 1) representation decorrelation to disentangle popularity from latent features [21, 27], 2) popularity-aware alignment and instance reweighting [1, 16], and 3) cluster-based knowledge distillation from popular to niche items [2]. While these methods collectively maintain recommendation accuracy by separating popularity signals from relevance, current approaches remain fundamentally constrained by their oversimplified bias-reduction frameworks. Key limitations include inadequate consideration of user heterogeneity in item perception and failure to account for dynamic exposure patterns in real-world platforms, ultimately preventing effective reconciliation of fairness objectives with recommendation quality.

Two-sided fairness-aware recommendation. Two-sided fairness requires recommender systems to simultaneously achieve utility parity across users and exposure equity across items. Current research primarily focuses on ensuring performance-based user fairness and exposure-based item fairness. Existing approaches can be categorized into two paradigms: 1) post-processing methods that employ greedy re-ranking framed as fair allocation problems [17] or combinatorial optimization [13], though these remain constrained by heuristic solutions and opaque biases inherited from pre-trained models [20]; and 2) multi-objective constrained optimization that incorporates fairness regularizers (e.g., group utility constraints [17]) through multi-task learning frameworks, which often struggle with conflicting objectives and unstable trade-off balancing [22]. Departing from these conventional approaches, we contend that addressing two-sided bias fundamentally requires investigating its underlying causal mechanisms.

3 Preliminaries

3.1 Problem Formulation

We formalize the top- k recommendation task with two-sided fairness considerations for both customers (users) and providers. Formally, let $U = \{u\}_{u=1}^m$ and $I = \{i\}_{i=1}^n$ denote the sets of m users and n items respectively, with historical interactions represented by a binary matrix $R \in \{0, 1\}^{m \times n}$ where $I_u = \{i \in I \mid R_{u,i} = 1\}$ indicates items interacted by user u . Each item i is associated with a provider $p_i \in P = \{p\}_{p=1}^l$, and we write $I_p \subseteq I$ for items belonging to provider p . The recommender system learns a scoring function $f_\theta : U \times I \rightarrow [0, 1]$ that predicts interaction probabilities $\hat{R}_{u,i} = f_\theta(u, i)$, ultimately generating personalized recommendation lists $L_u \subseteq I$ containing the top- k ranked items for each user u .

3.2 Personalized Brand Effect

Building upon established metrics of item popularity and user activity [7, 9], we first formalize these foundational concepts.

Definition 1 (Item Popularity, IP). For item $i \in I$, its popularity g_i is defined as the normalized user interaction frequency $g_i = |U_i|/|U|$, where $U_i = \{u \in U \mid R_{u,i} = 1\}$ denotes the set of users who interacted with i .

Definition 2 (User Activity, UA). For user $u \in U$, his/her user activity a_u measures engagement through item interaction count $a_u = \text{count}(I_u)$, where $\text{count}(I_u)$ is the number of items in I_u .

To capture the latent relationships between users, items and providers, we propose the concept of Personalized Brand Effect (PBE). This novel metric quantifies latent cross-entity influences by first establishing user-provider similarity.

Definition 3 (Users' Provider Similarity). Given two users $u, v \in U$, their provider-based similarity $\text{sim}(u, v)$ is defined through the Jaccard index of their interacted providers: $\text{sim}(u, v) = |P_u \cap P_v|/|P_u \cup P_v|$, where P_u (or P_v) denotes the set of providers associated with user u 's (or v 's) historical interactions. This metric quantifies behavioral similarity by measuring the overlap between providers, where higher values indicate stronger alignment in consumption preferences.

Definition 4 (Similar User Set). For a target user u , we define his/her similar user set $S_u^p \subseteq U$ as the top- k nearest neighbors under the users' provider similarity metric: $S_u^p = \arg \max_{v \in U \setminus \{u\}}^k \text{sim}(u, v)$, where k is a predefined neighborhood size hyper-parameter.

Definition 5 (Personalized Brand Effect, PBE). For each user-item pair (u, i) , the Personalized Brand Effect $pbe_{u,i} \in [0, 1]$ quantifies provider p_i 's influence within u 's interest community by computing the fraction of similar users (i.e., S_u^p) who have interacted with p_i 's items: $pbe(u, i) = |S_i^u|/|S_u^p|$ where $S_i^u = \{v \in S_u^p \mid \exists j \in I_{p_i}, R_{v,j} = 1\}$.

The PBE metric captures the strength of brand penetration in u 's social context. A higher value indicates stronger community endorsement. Specifically, $|S_i^u|$ reflects how deeply the products of brand p_i permeate u 's similarity neighborhood, while the denominator normalizes for user activity levels.

3.3 Preliminaries for Counterfactual Inference

3.3.1 Causal Graph. A causal graph is a directed acyclic graph (DAG), where the nodes represent random variables, and the edges represent causal relations. For example, $A \rightarrow L$ means that alcohol A has a direct effect on lung cancer L , and $A \rightarrow C \rightarrow L$ means that alcohol affects lung cancer indirectly through smoking C . The value of L can be computed as:

$$L_{a,c} = L(A = a, C = c), \quad C_a = C(A = a) \quad (1)$$

3.3.2 Counterfactual Inference. Counterfactual inference imagines outcomes from changing a variable's value. For example, Figure 2 illustrates the effect of alcohol consumption on lung cancer, where A , C , and L represent alcohol, cigarette, and lung cancer, respectively. In Figure 2(b), we ask what happens if alcohol consumption is set to a different value. Gray nodes represent reference states used to estimate causal effects [28]. The counterfactual scenario is shown in Figure 2(c), where C and L are set to counterfactual values.

3.3.3 Causal Effect. The causal effect of A on L is how much L changes when A changes. The total effect (TE) is:

$$TE = L_{a,c} - L_{a^*,c^*} \quad (2)$$

Here TE is decomposed into natural direct effect (NDE) and TIE :

$$NDE = L_{a,c^*} - L_{a^*,c^*}, \quad TIE = TE - NDE = L_{a,c} - L_{a,c^*} \quad (3)$$

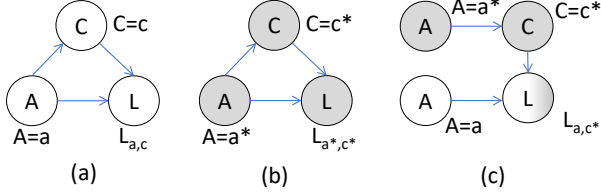


Figure 2: Illustration of causal graphs about the effect of alcohol consumption on lung cancer.

Table 1: Notation.

Symbols	Definitions and Descriptions
u, i, p	User u , Item i , Provider p
$e_{u/i}$	ID-based embedding of user u / item i
$\Pi_{i/u}$	The system prompt
$Q_{u/i}$	User / Item input prompt
$P_{u/i}$	User / Item profile
$S_{x,u,i}$	Factual prediction score under the current values of X , U , and I
S_{x^*,u^*,i^*}	Counterfactual prediction score where U and I are set to u^* and i^*
S_{x^*,u^*,i^*}	Counterfactual prediction score where X , U , and I take values x^* , u^* , and i^*
S_{fair}	Fair prediction score
$\hat{x}_{u,i}$	Confounding effect estimation involving IP , UA , and PBE
$\hat{g}_{u,i}$	Predicted interaction probability between user u and item i
$\hat{g}_{i,u}, \hat{u}_a, \hat{p}be_{u,i}$	Predicted IP/UA (for i/u) and PBE (for user u and item i)
$g_{i,u}, u_a, pbe_{u,i}$	The observed IP , UA , PBE calculated on the training set
$\hat{r}_{u,i}$	Base model prediction for user u and item i
γ, δ, β	Tunable weighting coefficients for PBE , IP , and UA

4 Our Proposed Model: LeadFairRec

In this work, we propose a novel model (i.e., LeadFairRec) for two-sided fairness-aware recommendation, as shown in Figure 4. The notations used in this paper are listed in Table 1. Our model comprises three core components: two-sided causal graph construction, discriminative counterfactual debiasing, and LLM-enhanced counterfactual inference. First, we use the two-sided causal graph construction component to jointly model provider-customer fairness interactions through their causal relationships. Second, based on the constructed two-sided causal graph, the discriminative counterfactual debiasing component removes spurious correlations while maintaining true user-item interactions. Finally, in order to enhance the robustness of causal debiasing, the LLM-enhanced counterfactual inference component uses LLMs (DeepSeek) to semantically enrich user/item profiles via contrastive learning. We will detail these components in this section.

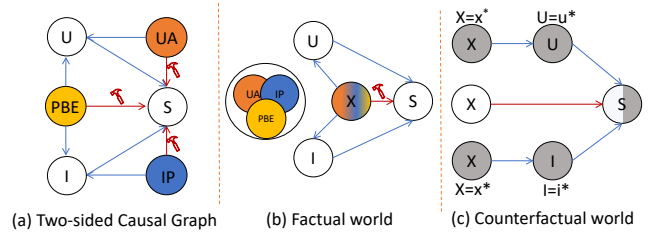


Figure 3: Illustration of two-sided causal graphs.

4.1 Two-sided Causal Graph Construction

To operationalize the two-sided bias phenomenon, we construct a two-sided causal graph (Figure 3a) that captures both isolated and joint bias effects. The graph structure represents core entities as endogenous variables: user representations (U), item representations (I), and prediction scores (S), while explicitly modeling key bias factors - user activity (UA) and item popularity (IP) - as exogenous variables that independently influence U and I respectively. Crucially, we introduce PBE as a quantitative mediator that captures triadic dependency structures among users, items, and providers.

From a causal inference perspective, our analysis reveals three distinct bias propagation pathways in recommender systems: 1) High- IP items benefit from popularity-exposure feedback loops where initial popularity drives increased visibility, creating a self-reinforcing cycle; 2) High- UA users receive preferential treatment through engagement-driven feedback mechanisms that amplify their exposure in recommendation results; and 3) High- PBE items gain advantage from community endorsement effects, where strong brand penetration within user similarity clusters signals quality to the system. These interconnected causal pathways ($UA \rightarrow S$, $IP \rightarrow S$, and $PBE \rightarrow S$) systematically distort prediction scores S by introducing confounding effects that require targeted intervention through counterfactual debiasing strategies to isolate and remove their spurious influence while preserving genuine user preferences. The systematic construction of a causal graph in an emerging domain entails the identification of core entities, the definition of deviation factors, the simulation of exogenous variables, the establishment of causal pathways, and the analysis of deviation propagation.

4.2 Discriminative Counterfactual Debiasing

Building on the insight that certain biases contain valuable signals, we propose a discriminative counterfactual debiasing approach that strategically preserves beneficial bias components while eliminating unfair influences through a three-stage causal framework: 1) We first introduce a composite proxy variable X that systematically integrates UA , IP , and PBE within a unified causal graph (Figure 3b), capturing multi-source confounding effects; 2) Building on counterfactual inference (Section 3.3), we then decompose prediction scores into Total Effect ($TE = S_{x,u,i} - S_{x^*,u^*,i^*}$), Natural Direct Effect ($NDE = S_{x,u^*,i^*} - S_{x^*,u^*,i^*}$), and Total Indirect Effect ($TIE = TE - NDE$) components, where $S_{x,u,i}$ represents the factual prediction, S_{x,u^*,i^*} denotes counterfactual counterpart under reference states (u^*, i^*), and S_{x^*,u^*,i^*} denotes counterfactual counterpart under reference states (x^*, u^*, i^*) (Figure 3c); 3) The

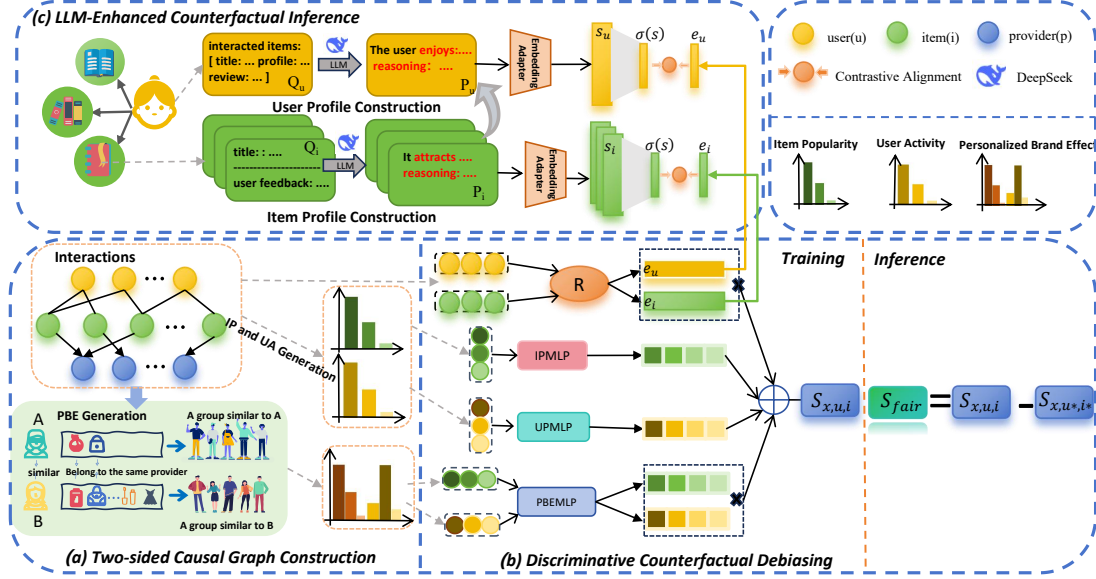


Figure 4: Framework of LeadFairRec Model.

final debiasing operation isolates *TIE* as the fairness-adjusted score $S_{fair} = S_{x,u,i} - S_{x,u^*,i^*}$. This discriminative preservation of *TIE* effectively maintains valuable signals while removing spurious correlations through causal isolation.

The factual score $S_{x,u,i}$ combines a recommendation model f_R (here it can be an arbitrary recommendation model such as matrix factorization, neural networks, or graph-based approaches) with a composite confounder $\hat{x}_{u,i}$:

$$S_{x,u,i} = \hat{x}_{u,i} \cdot f_R(U = u, I = i) \quad (4)$$

$$\hat{x}_{u,i} = \sigma(f_{PBE}(U = u, I = i)) \times \sigma(f_{UA}(U = u)) \times \sigma(f_{IP}(I = i)) \quad (5)$$

where σ denotes sigmoid activation and f are specialized bias estimators.

The counterfactual score S_{x,u^*,i^*} explicitly blocks direct user/item influence paths through reference states (u^*, i^*) while maintaining the learned weights of three key bias components - *PBE* (γ), *IP* (δ), and *UA* (β). This design ensures that $f_R(U = u^*, I = i^*)$ operates as a path-blocking constant while maintaining pair-specific confounder effects through $\hat{x}_{u,i}$, allowing the precise isolation of indirect effects.

$$S_{x,u^*,i^*} = \hat{x}_{u,i} \cdot f_R(U = u^*, I = i^*) = \gamma \cdot pbe_{u,i} + \delta \cdot g_i + \beta \cdot a_u \quad (6)$$

where g_i is the observed *IP* of item i , a_u is the observed *UA* of user u , and $pbe_{u,i}$ is the observed *PBE* for user u and item i , all evaluated on the training dataset.

4.3 LLM-Enhanced Counterfactual Inference

To alleviate the noisy effects in the learned representations, we propose an LLM-enhanced counterfactual inference strategy. First,

we leverage LLM's reasoning capabilities to generate discriminative user/item profiles. Then we establish contrastive alignment between traditional ID embeddings (e) and LLM-generated semantic embeddings (s). This strategy mitigates spurious correlations while preserving genuine interaction patterns.

4.3.1 Profile Generation via Reasoning. The profile generation process employs a structured prompting architecture implemented through Deepseek, formally expressed as $P_u = \text{LLM}(\Pi_u, Q_u)$ for users and $P_i = \text{LLM}(\Pi_i, Q_i)$ for items, where $\Pi_{u/i}$ represents the carefully designed system prompts that provide contextual guidelines to LLM, and $Q_{u/i}$ denotes the specific query templates for user and item profile generation, respectively. By strategically combining these system prompts with the targeted profile generation queries, the framework produces enhanced textual profiles P_u (user profiles) and P_i (item profiles) that more accurately capture the nuanced characteristics and interaction patterns of users and items in the recommender system.

4.3.2 Contrastive Alignment. The semantic alignment process begins by encoding the generated user/item profiles (P_u or P_i) into dense vector representations through a pretrained text embedding model $T(\cdot)$, resulting in semantic embeddings $s_u = T(P_u)$ and $s_i = T(P_i)$. These semantic representations are then aligned with their corresponding ID embeddings (e_u or e_i) via a contrastive learning framework. Inspired by the work [18], we formulate the alignment using a density ratio function $f(s_i, e_i)$ that measures the similarity between semantic and ID representations:

$$f(s_i, e_i) = \exp(\text{sim}(\sigma_{\downarrow}(s_i), e_i)) \quad (7)$$

where $\text{sim}(\cdot)$ denotes cosine similarity, $\sigma_{\downarrow}$ represents a multi-layer perceptron that projects the semantic representation s_i into the

feature space of e_i , and the exponential function transforms the similarity score into a positive real value. During contrastive learning, the pairs (s_i, e_i) are treated as positive samples. The model learns to pull positive pairs closer and pull apart negative pairs.

4.4 Training and Inference

4.4.1 Training Phase. Based on the aforementioned estimation framework, we jointly train f_R , f_{PBE} , f_{UA} and f_{IP} . The primary objective during training is for the model to learn and adapt to the inherent biases present in the training data, rather than directly generating user recommendations. This necessitates a distinct estimation approach for interaction probabilities that differs from our inference methodology. The training set is constructed according to the causal graph representation of the factual world (Figure 3(a)), where all causal effects remain unconstrained. To estimate the interaction probability $\hat{y}_{u,i}$ between user u and item i in historical training data, we follow the estimation procedure of score $S_{x,u,i}$. This approach assumes that all causal paths in Figure 3(a) actively contribute to the final prediction:

$$\begin{aligned}\hat{y}_{u,i} &= \hat{x}_{u,i} \times f_R(U = u, I = i) \\ &= \sigma(p\hat{b}e_{u,i}) \times \sigma(\hat{a}_u) \times \sigma(\hat{g}_i) \times \hat{r}_{u,i}\end{aligned}\quad (8)$$

where $\hat{r}_{u,i} = f_R(U = u, I = i)$ denotes the prediction of the base model, $\sigma(p\hat{b}e_{u,i}) = \sigma(f_{PBE}(U = u, I = i))$, $\sigma(\hat{a}_u) = \sigma(f_{UA}(U = u))$ and $\sigma(\hat{g}_i) = \sigma(f_{IP}(I = i))$ represent the estimated PBE , UA and IP values respectively.

We optimize the model parameters using the Bayesian Personalized Ranking (BPR) loss [6, 15, 19], a widely adopted objective function in recommender systems:

$$L_R = \sum_{(u, i^+, i^-) \in O} -\ln \sigma(\hat{y}_{u,i^+} - \hat{y}_{u,i^-}) \quad (9)$$

where $O = \{(u, i^+, i^-) \mid (u, i^+) \in R^+, (u, i^-) \in R^-\}$ represents the training set, R^+ denotes the set of observed positive interactions, i^+ indicates an item with which user u has actually interacted, R^- contains randomly sampled negative interactions and i^- represents a randomly selected item that u has never interacted with.

We train bias estimation models f_{PBE} , f_{UA} , f_{IP} as regression tasks, using observed behavioral data as ground truth. Following established practices in causal recommendation [20, 24], we employ Mean Squared Error (MSE) loss functions for optimization. For f_{PBE} , the loss value L_{PBE} is $\frac{1}{|R|} \sum_{(u,i) \in R} (pbe_{u,i} - \sigma(p\hat{b}e_{u,i}))^2$. L_{UA} and L_{IP} are defined analogously.

To maximize mutual information between semantic and collaborative representations, we formulate the following contrastive alignment objective:

$$L_{info} = \mathbb{E} \log \left[\frac{f(s_i, e_i)}{\sum_{s_j \in S} f(s_j, e_i)} \right] \quad (10)$$

The complete optimization objective combines all loss components through a weighted formulation:

$$L = L_R + \alpha(L_{PBE} + L_{IP} + L_{UA}) - \epsilon L_{info} + \lambda \|\Theta\|_2^2 \quad (11)$$

where α balances the bias estimation losses, ϵ controls the mutual information maximization, and λ regulates the L_2 -regularization on all trainable parameters Θ .

4.4.2 Inference Phase. Rather than relying on predictions from the factual world, we mitigate recommendation bias by removing the NDE and applying Eq. 6 for recommendations. By incorporating these prior estimates, we derive the following inference formulation:

$$\tilde{y}_{u,i} = \hat{y}_{u,i} - S_{x,u^*,i^*} \quad (12)$$

During the ranking adjustment phase, the final ranking score for each candidate item is computed using Eq. 12.

5 Experiments

In this section, we perform experiments to demonstrate the effectiveness of LeadFairRec. We aim to answer the following research questions. RQ1: Does the proposed LeadFairRec outperform other state-of-the-art methods for fairness recommendation? RQ2: Does the proposed LeadFairRec benefit from our well-designed components? RQ3: How do the key hyper-parameters impact the performance of LeadFairRec? RQ4: How does our LeadFairRec perform w.r.t training efficiency?

5.1 Experimental Setup

5.1.1 Dataset Descriptions. We select three public datasets, i.e., *BookCrossing* [32], *AmazonVideoGames* [14], and *BeerAdvocate* [11] to conduct experiments considering typical domains (books, games, and beverages) and different scales. To evaluate the two-sided fairness, publishers in the book domain, companies in the game domain, and brewers in the beer domain are extracted from attribute information as the provider of items, which is consistent with actuality. For datasets, we filter users and items with less than five interactions. The statistics of the three preprocessed datasets are shown in Table 2.

Table 2: Dataset Statistics

Dataset	#Users	#Items	#Interactions	#Providers
BookCrossing	6400	8496	106918	607
AmazonVideoGames	44189	14752	394832	1621
BeerAdvocate	14041	20562	1361000	2694

5.1.2 Evaluation Metrics. We choose the widely used $NDCG@k$ as the metric ($k = 20$ as applied in [6, 10]) to evaluate the accuracy of recommendation, and P-Fair and C-Fair are utilized to measure two-sided fairness. Referring to previous studies, we will now give the specific definitions of P-Fair and C-Fair.

$$\text{P-Fair} = cv(e_{p_1}, e_{p_2}, \dots, e_{p_l}), \quad (13)$$

$$\text{C-Fair} = cv(NDCG_{u_1}, NDCG_{u_2}, \dots, NDCG_{u_m}). \quad (14)$$

The exposure of an item l , denoted as e_{p_l} , depends on its position in the top- k list. The exposure of a provider is then defined as the average exposure score of all items it owns. $NDCG_{u_m}$ represents the NDCG for user m . The coefficient of variation, denoted $cv(\cdot) = \frac{\sigma(\cdot)}{\mu(\cdot)}$, is used to measure the dispersion of exposure scores within a given set. A lower coefficient of variation indicates a higher level of fairness in the exposure distribution.

To evaluate the overall performance of various metrics, a common solution is to define the harmonic average of accuracy and other metrics (e.g., diversity) as the F-score [3, 10]. Similarly, we use an AF-score indicator based on the trade-off between bilateral fairness and accuracy. To facilitate the comparison of baselines, we define the AF-score to measure comprehensive performance within three metrics [3, 10]:

$$AF = \frac{NDCG}{P\text{-Fair} \times C\text{-Fair}} \quad (15)$$

The higher the AF-score, the better the overall performance in terms of accuracy and two-sided fairness.

5.1.3 Baselines. We compare LeadFairRec with baselines:

- **Original** [6, 19]: The default setting for training regardless of fairness.
- **PPAC** [16]: A method for debiasing recommendation with personal popularity.
- **CPFair** [13]: Three post-processing algorithms aimed at achieving two-sided fairness.
- **RLMRec** [18]: Representation learning with large language models for recommendation.
- **Ada2Fair** [25]: Promoting two-sided fairness with adaptive weights for providers and customers in recommendation.

5.1.4 Implementation Details. We implement all models based on the open-source benchmark library RecBole [26, 29, 30]. Since our method is model-agnostic, we select two representative methods, BPRMF [19] and LightGCN [6], as base recommenders. For the general parameters of all methods, we tune the learning rate in $\{1e-4, 5e-4, 1e-3, 5e-3, 1e-2\}$ and the number of graph convolutional layers in $\{1, 2, 3\}$. For LeadFairRec, we set the *PBE* coefficient $\gamma = -256$, the *IP* coefficient $\delta = 64$, and the *UA* coefficient $\beta = 1$. In the LLM-enhanced causal inference module, we utilize the DeepSeek model and BERT to generate semantic representations for user and item profiles. For baselines, we carefully search for hyperparameters following the settings of the original authors.

5.2 Overall Performance: RQ1

Table 3 presents a comparative performance analysis of LeadFairRec against various baseline models among different base recommenders, evaluated using the C-Fair, P-Fair, NDCG@20, and AF metrics. We note the following key observations:

Comparison with Single-sided Fairness Models. The PPAC model concentrates on item popularity to improve accuracy by correcting for exposure bias. Despite improving accuracy and moderately reducing item-side exposure disparities, its effect on item-side fairness is constrained, as it fails to account for user-side fairness and the provider-fairness implications of item popularity. This underscores the limitations of single-sided fairness models. Conversely, our proposed LeadFairRec model adopts a two-sided fairness strategy, simultaneously addressing both user and provider fairness, thereby effectively alleviating unfairness from dual perspectives.

Comparison with Two-sided Fairness Models. The Ada2fair and CFPfair aim to enhance two-sided fairness. Experimental results indicate that in all three datasets, both achieve improvements

in two-sided fairness compared to baseline models. However, compared with our proposed method, these two models still lag in the continuous optimization of fairness. A deeper analysis suggests that their lack of a thorough causal analysis of model unfairness and the failure to incorporate semantic information enhancement might be the key factors restricting their fairness-improvement effectiveness.

Contrasting LeadFairRec with LLM-Enhanced Recommendation Baselines. The RLMRec achieves a significant boost in accuracy. The enriched semantic information on both the user and item sides provided by the LLMs also helps alleviate user-side and provider-side unfairness to some extent across certain datasets. However, fairness was not the core optimization goal in this model’s design, and its performance in two-sided fairness metrics remains suboptimal. However, this approach offers valuable insights and directions for improving two-sided fairness. In contrast, our proposed LeadFairRec model also integrates LLM to enhance causal inference. Unlike this model, LeadFairRec not only boosts accuracy but also achieves a more optimal performance in two-sided fairness metrics. This is attributed to LeadFairRec’s comprehensive consideration of fairness in its core design and optimization process.

Superior Comprehensive Performance of LeadFairRec. Based on the experimental results, our proposed LeadFairRec model demonstrates exceptional effectiveness in mitigating two-sided unfairness, delivering marked improvements over baseline models. In terms of accuracy, it matches or surpasses other models. In comprehensive metrics balancing accuracy and fairness, LeadFairRec excels, particularly in the AF Score, which gauges this balance. A higher AF Score signifies better performance, and as shown in Figure 5, LeadFairRec exhibits superior AF-Score performance across multiple datasets and backbone networks.

5.3 Ablation Study: RQ2

To gain deeper insights into the design of LeadFairRec, we conduct ablation studies by disabling specific components. Specifically, we develop four variants:

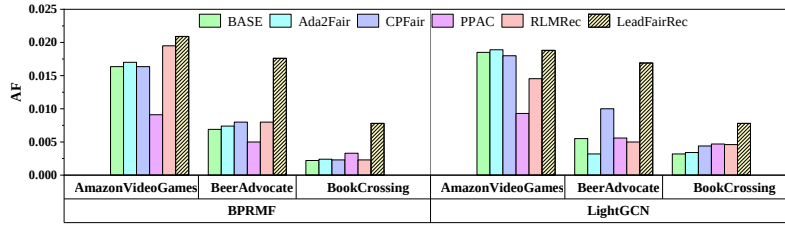
- (1) **w/o LLM** ignores LLM-Enhanced Counterfactual Inference and removes all components related to semantic information.
- (2) **w/o PBE** removes the *PBE* estimation model, denoted as f_{PBE} , along with all related *PBE* terms during both the training and inference phases.
- (3) **w/o IP** removes the *IP* estimation model, denoted as f_{IP} , along with all related *IP* terms during both the training and inference phases.
- (4) **w/o UA** removes the *UA* estimation model, denoted as f_{UA} , along with all related *UA* terms during both the training and inference phases.

Figure 6 demonstrates that the complete LeadFairRec architecture achieves significantly higher AF scores compared to any ablated variant. This consistent performance advantage across all datasets provides empirical evidence for the complementary value of each design component in our model.

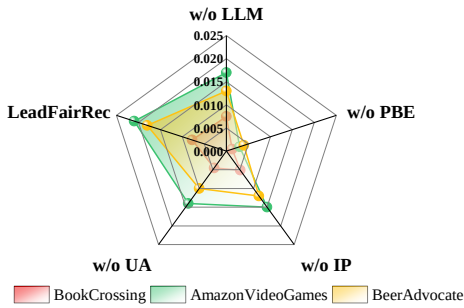
Figure 7 presents a fine-grained analysis of how critical model components individually influence various evaluation metrics: 1) removing LLM-enhanced semantic alignment degrades both C-Fair and P-Fair, highlighting its crucial role in noise-resistant fairness

Table 3: Overall performance comparison (best results in bold, second best underlined).

Dataset		AmazonVideoGames				BeerAdvocate				BookCrossing			
Backbone	Methods	C-Fair ↓	P-Fair ↓	NDCG ↑	AF ↑	C-Fair ↓	P-Fair ↓	NDCG ↑	AF ↑	C-Fair ↓	P-Fair ↓	NDCG ↑	AF ↑
LightGCN	Base	2.1819	3.2085	0.1297	0.0185	1.9003	7.8582	0.0823	0.0055	3.2188	4.8339	0.0498	0.0032
	+Ada2Fair	2.2014	<u>3.1332</u>	0.1304	0.0189	2.2455	11.6345	0.0858	0.0033	3.1832	4.5621	0.0502	0.0034
	+CFFair	2.0843	3.4751	<u>0.1304</u>	0.0180	1.5863	<u>6.6663</u>	<u>0.1066</u>	<u>0.0100</u>	2.9136	4.1659	0.0544	0.0044
	+PPAC	2.2174	4.9105	0.1023	0.0093	1.8923	9.8976	0.1065	0.0056	3.1768	4.1562	0.0624	<u>0.0047</u>
	+RLMRec	2.1518	3.7479	0.1172	0.0145	1.8032	10.0943	0.0915	0.0050	3.0350	3.8298	0.0542	0.0046
	+LeadFairRec (ours)	2.2074	2.9105	0.1231	<u>0.0188</u>	1.4423	6.4573	0.1578	0.0169	<u>2.9333</u>	2.6808	<u>0.0615</u>	0.0078
BPRMF	Base	2.2740	3.3852	0.1259	0.0163	1.8246	6.8868	0.0878	0.0069	3.5742	5.1103	0.0411	0.0023
	+Ada2Fair	2.2421	<u>3.0342</u>	0.1159	0.0170	1.7956	6.7065	0.0893	0.0074	3.6432	<u>4.4943</u>	0.0398	0.0024
	+CFFair	2.1492	3.6982	0.1300	0.0163	1.6481	7.2026	<u>0.0952</u>	0.0080	3.2501	5.8511	0.0452	0.0023
	+PPAC	2.6827	3.7349	0.0914	0.0091	1.9270	8.4846	0.0822	0.0050	<u>3.1634</u>	5.7640	<u>0.0607</u>	<u>0.0033</u>
	+RLMRec	2.1540	3.0418	<u>0.1279</u>	<u>0.0195</u>	1.7889	<u>6.4197</u>	0.0919	<u>0.0080</u>	3.3061	5.6606	0.0441	0.0023
	+LeadFairRec (ours)	2.6178	2.0224	0.1108	0.0210	1.5301	5.0341	0.1361	0.0176	2.9315	2.6921	0.0614	0.0078

**Figure 5: Comparison of AF-score for different methods.**

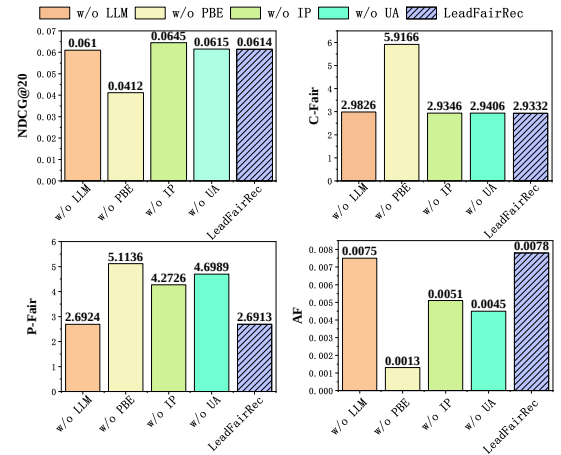
modeling; 2) removing the debiasing procedure for the joint factor *PBE* results in a consistent decline across all evaluation metrics, demonstrating its critical role in preserving both two-sided fairness (C-Fair, P-Fair) and recommendation accuracy (NDCG); and 3) removing *IP* debiasing significantly degrades P-Fair, while excluding *UA* debiasing compromises both C-Fair and NDCG, which collectively reveals cross-sided fairness interdependence and underscores the necessity of holistic fairness in recommendation systems.

**Figure 6: Impact of different variants on AF.**

5.4 Impact of Hyper-parameters: RQ3

In this experiment, we conduct a systematic study on the influence of hyperparameters, specifically the *PBE* coefficient γ , the *IP* coefficient δ , and the *UA* coefficient β , on the performance of the LeadFairRec model. By evaluating these parameters across three

distinct datasets, we aim to identify the optimal configuration that enhances the model's effectiveness and fairness, ensuring robust and reliable recommendations. This experiment focuses on the issue of fairness optimization in recommendation systems, with experimental designs carried out from two perspectives: single factor (*IP* or *UA*) and joint factor (*PBE*). In the single factor perspective exper-

**Figure 7: Performance of variants on the BookCrossing dataset in terms of NDCG, P-Fair, C-Fair, and AF.**

iment, we fixed other hyperparameters (e.g., $\gamma = -256$) and adjusted β and δ by selecting values from the set {16, 32, 64, 128, 256} to evaluate their impact on the C-Fair and P-Fair metrics. The results,

as shown in Figure 8, indicate that C-fair is optimized when $\beta = 128$ and $\delta = 16$, while P-fair performs best when $\beta = 16$ and $\delta = 32$. This indicates that β and δ play a key role in adjusting the weights of item (IP) and user (UA) features. Moreover, the varying effects of hyperparameter combinations across different datasets reveal the heterogeneous nature of unfairness phenomena within each dataset. The inherent biases and feature distributions can differ significantly from one dataset to another. For instance, recruitment datasets may exhibit gender - specific preferences, whereas educational datasets might show regional student-specific biases. These issues can be addressed through adjustments to the hyperparameters.

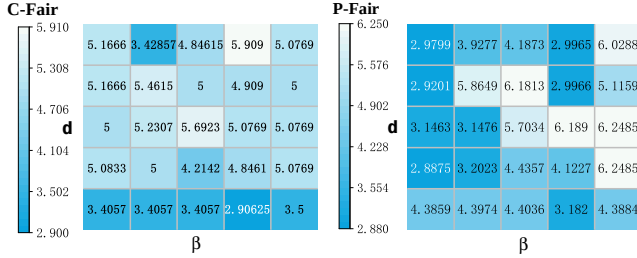


Figure 8: Impact of Hyperparameters β and δ on C-Fair and P-Fair

In the subsequent experiment, we delved into the impact of the two-sided causal factor *PBE* parameter γ on model accuracy, fairness, and overall performance from a combined perspective. We maintained β and δ at constant values and varied γ across the range of $\{-256, -128, -64, -32\}$. As depicted in the Figure 9, γ influences C-Fair, P-Fair, NDCG, and AF in a non-linear manner. While C-Fair prefers the lower end of the γ spectrum, P-Fair and NDCG peak at $\gamma = -128$. The AF metric shows a separate optimum at $\gamma = -64$. This experiment underscores the significance of γ in fine-tuning the balance between model accuracy and fairness. Choosing an appropriate γ value is vital for aligning model performance with specific application requirements. This finding further enriches our understanding of how hyperparameters can be strategically adjusted to enhance model performance across multiple dimensions.

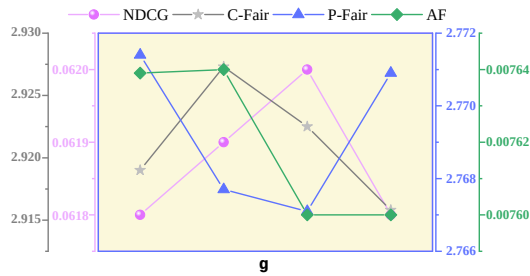


Figure 9: Impact of Parameter γ on Model Metrics

5.5 Analysis of Training Efficiency: RQ4

We conduct a thorough analysis of the time complexity associated with the LeadFairRec model, focusing on its core components. The

causal graph construction (Section 4.1) is a preliminary step that does not directly influence the training time complexity. And the calculation of both the factual score $S_{x,u,i}$ and the counterfactual score S_{x,u^*,i^*} (Section 4.2) involves Multi-Layer Perceptron (MLPs), each with a complexity of $O(N \times d_s \times d_e)$, where N denotes the batch size, and d_s and d_e represent the dimensions of user and item features, respectively. The decomposition into TE, NDE, and TIE, although involving additional steps, remains linear and does not significantly increase the complexity beyond $O(N \times d_s \times d_e)$. The generation of profiles using an LLM (Section 4.3) is identified as computationally intensive, with each profile complexity assumed to be $O(L \times d)$, where L is the sequence length and d is the embedding dimension. Consequently, the total complexity for generating profiles for all users and items is $O((U + I) \times L \times d)$, where U and I are the counts of users and items, respectively. The contrastive alignment, which computes similarities between semantic and ID embeddings, adds $O(N \times d)$ complexity per batch. The overall complexity of the training is dominated by operations within neural networks and the LLM, which are the most computationally intensive parts. The complexity per epoch can be approximated as $O(B \times N \times (d_s \times d_e + L \times d))$, where B is the number of batches per epoch, N is the batch size, $d_s \times d_e$ is the complexity of MLP operations for user and item representations, and $L \times d$ is the complexity of LLM operations for profile generation. For larger models with improved performance, the additional time complexity is only about 10% to 20% compared to the original time.

6 Conclusion

In this work, we propose LeadFairRec, a novel model that In this paper, we addressed the significant challenge of achieving multi-stakeholder fairness in recommender systems. We proposed the LeadFairRec model, which is designed to balance user and item fairness while preserving recommendation accuracy. First, we construct a two-sided causal graph that formally quantifies the inter-dependent causal pathways between user satisfaction equity and provider exposure fairness. Second, we develop a discriminative counterfactual debiasing mechanism, which identifies and removes spurious correlations via path-specific effect estimation, and use a benefit-aware preservation strategy that retains causally grounded preference signals. Third, we introduce an LLM-enhanced counterfactual inference engine that generates noise-resilient user/item representations to significantly enhance the fidelity and robustness of causal effect estimation. Our experimental results on three real-world datasets demonstrate the effectiveness of LeadFairRec in enhancing both fairness and accuracy. For future work, we plan to explore more sophisticated causal inference techniques and investigate the mitigation of discrimination arising from user-sensitive attributes (e.g., gender, age, ethnicity) by developing attribute-aware fairness constraints and privacy-preserving representation learning techniques.

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GenAI Usage Disclosure

In our research, we utilized large language models (LLMs) to refine and revise sentences across the entire manuscript. These tools were applied to enhance the quality of each section and improve textual clarity, ensuring the content is meticulously crafted and free of linguistic inaccuracies.

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